| NOT MEASUREMENT |
| SENSITIVE |

MIL-H-21103E
24 May 1989
SUPERSEDING
MIL-H-21103D
25 February 1983

MILITARY SPECIFICATION

HAMMER, PILEDRIVER, SELF-POWERED, DIESEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-powered, diesel piledriver hammers. The hammers are intended for use in driving wood, steel pipe, steel wide flange beam, concrete, and sheet piling.

1.2 Classification. The hammers will be one of the following types and sizes, as specified (see 6.2):

Type I - For use with 20- to 22-inch leads.

Size A - 8,100 foot-pounds (ft-lb) (minimum) striking energy.
Size B - 16,000 ft-lb (minimum) striking energy.
Size C - 22,400 ft-lb (minimum) striking energy.
Size D - 8,000 ft-lb (minimum) striking energy, 19 horsepower (hp).
Size E - 18,000 ft-lb (minimum) striking energy, 28 hp.

Type II - For use with 26-inch leads.

Size F - 30,000 ft-lb (minimum) striking energy.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

1.2.1 Part or Identifying Number (PIN). The PIN for items described in this specification will be identified as shown in section 6.5.

M21103 - XX - XX

Specification number
Type code number (see 6.5.2)
Size code number (see 6.5.3)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape, Packaging, Waterproof.
PPP-T-76 - Tape, Pressure Sensitive Adhesive, Packaging/Paper.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproof, Waterproofed, Flexible.
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Material.
MIL-C-3774 - Crate, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-C-20218 - Chromium Plating, Electro-Deposited, Porous.
MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat Sealable.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-595 - Colors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Water-proofing; with Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 328 - Standard Specification for Steel Sheet Piling. (DoD adopted)
ASTM D 3951 - Commercial Packaging, Practice for. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Structural Welding Code-Steel. (DoD adopted)

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJune Road, P.O. Box 351040 Miami, Florida 33135.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings. (DoD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The hammer shall be self-contained, consisting essentially of a cylinder, free piston or ram, anvil, starting and stopping devices, fuel and lubricating oil systems, and throttle control. The self-powered hammer shall be air-cooled and shall operate on the two-stroke cycle compression-ignition principle. The upper end of the cylinder may be either open or closed. The open-end cylinder-type hammer shall depend on gravity to arrest the upward stroke of the ram. The closed-end cylinder-type hammer shall be equipped with a compression tank that shall assist the force of gravity in arresting the upward stroke of the ram. The hammer shall be furnished with lead guide, drive caps, leads, and bottom braces as specified in 3.11, 3.19, and 3.19.1.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.2.1.

3.3 Standard commercial product. The piledriver shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the piledriver being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Safety. Reciprocating parts or components subjected to high temperatures, which are of such a nature or so located as to become a hazard to operating personnel, shall be enclosed, insulated, or guarded to the extent necessary to eliminate the hazard. This requirement does not apply to

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

necessary to eliminate the hazard. This requirement does not apply to air-cooling fins. Protective devices shall not impair the operating functions of the hammer. A device shall be provided that shall prevent accidental engagement of the starting mechanism during hammer operations. The open-end cylinder-type hammer shall be equipped with a retainer to prevent the ram from jumping out of the cylinder on the upward stroke. A transport plug shall be furnished with each open-end cylinder-type hammer. Plug shall secure the ram within the cylinder.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Design. The hammer shall be designed to incorporate devices that shall protect moving parts from inclement weather and abrasive particles. A cushioning device shall be designed and furnished with the hammer that shall isolate excessive vibrations resulting from the ram striking the anvil during piledriving operations. Design of the hammer shall permit ready accessibility to all parts for maintenance and repair in the field by use of conventional general-purpose tools associated with hammers. The design of a single hammer may incorporate features that shall permit the use of more than one weight of ram in the same cylinder.

3.8 Threads and fastening devices. All screw threads shall be in accordance with FED-STD-H28. Pins, screws, and bolts shall be installed with a positive locking method. Locking shall be by cotter pins, locknuts, castellated nuts, lockwashers, or other similar locking methods. Swaging, peening and staking of threads shall not be acceptable.

3.9 Operating weight, striking energy, and horsepower. The operating weight shall include the complete hammer, lead guides, heaviest drive cap fillers, and adapters to be used. The striking energy of each hammer shall be determined with the ram operating at maximum stroke (see 4.4.1.3) and at a cyclic rate of not less than 36 strokes per minute. The striking energy shall be calculated as follows:

a. Open-end cylinder-type.

$$E = RH$$

Where E = Striking energy in ft-lb
R = Weight of ram in pounds
H = Length of free-fall stroke in feet.

b. Closed-end cylinder-type.

(1) Positive pressure, scavenged type.

$$E = RH + \left[\frac{P_2V_2 - P_1V_1}{4.8} - \frac{P_1(V_1 - V_2)}{12}\right] - \left[\frac{P_3V_4 - P_1V_3}{4.8} - \frac{P_1(V_3 - V_4)}{12}\right]$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

(2) Atmospheric pressure, scavenged type.

$$\frac{P_3V_4 - P_1V_3}{4.8} - \frac{P_1(V_3 - V_4)}{12} = 0$$

Where E = Maximum striking energy in ft-lb
R = Weight of ram in pounds
H = Length of free-fall stroke in feet
P_1 = Atmospheric pressure in pounds per square inch (psi)
P_2 = Absolute pressure in bounce chamber and compressor tank at top of stroke psi (gauge plus atmosphere pressure)
P_3 = Absolute pressure in receiver after scavenge stroke and before opening air intake ports to power cylinder in psi
V_1 = Volume of bounce chamber (including compressor tank) above ram when ram closes atmospheric vent, measured in cubic inches (cu in.)
V_2 = Volume of bounce chamber (including compressor tank) above ram when ram is at highest point of stroke, measured in cu in.
V_3 = Volume of air receiver (including both side tanks and volume of annular scavenge cylinder under ram) when ram closes scavenge inlet port, measured in cu in.
V_4 = Volume of air receiver (including both side tanks and volume of annular scavenge cylinder under ram) at bottom of stroke, measured in cu in.

Horsepower shall be calculated as follows:

$$HP = \frac{E \times BPM}{33,000}$$

Where HP = Horsepower
E = Maximum striking energy in ft-lb
BPM = Cyclic rate in blows per minute.

Each size of hammer shall conform to the requirements specified in table I.

TABLE I. Operating weight, striking energy, and hp.

Type	Classification size	Operating weight (lb) (max)	Ram weight (lb) (min)	Striking energy (ft-lb) (min)	HP (min)
I	A	5,500	1100	8,100	
	B	7,500	2000	16,000	
	C	10,500	2800	22,400	
	D	6,000	1100	8,000	19
	E	13,000	2800	18,000	28
II	F	17,500	5000	30,000	

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

3.10 Performance. The hammer shall operate in any ambient temperature from -25° Fahrenheit (F) through +125°F. When operating with leads as specified herein, the hammer shall drive the piles specified in table II. While operating at maximum striking energy, the hammer shall drive a pile at refusal for a period of not less than 5 minutes without damage to the hammer. A pile that is driven not more than 1 inch in 10 blows while the hammer is operating at maximum striking energy, is defined as a pile driven to refusal. The hammer shall operate for not less than 50 hours without overheating, any component breaking or deforming permanently, fuel and lubrication leakage, excessive wear, and loss of adjustment of any component beyond the limits recommended in the manufacturer's maintenance manual. Each hammer shall demonstrate its operational capability for not less than 45 minutes.

3.11 Lead guides. Unless otherwise specified (see 6.2), the hammer shall be furnished with one set of lead guides. The guides shall allow the hammer to operate with commercial standard box-type piledriver leads.

3.12 Drive caps. Each hammer shall be furnished with universal or separate drive caps as specified (see 6.2). The drive caps shall be suitable for use with commercial standard box leads and for driving the piles listed in table II. Overlap between each drive-cap guide and the lead shall be not less than 2 inches. Hammers shall be furnished with a cushion block for inserting between the anvil and drive cap. The cushion block shall be either the disk type with laminated plastic and aluminum or the cast nylon type. The required type and quantity of cushion block to be furnished with each hammer shall be as specified (see 6.2).

3.13 Anvil, ram, and cylinder. The anvil shall be constructed of high quality forged steel or cast nodular iron. The ram shall be of forged or cast steel or nodular iron. The ram shall be equipped with compression rings located near its upper and lower ends for the closed-end cylinder-type hammer and near its lower end for the open-end cylinder-type hammer. If ram and cylinder are of dissimilar materials (i.e., nodular iron and steel), plating shall not be required. If ram and cylinder are of similar materials (i.e., carbon steel and alloy steel), plating of the cylinder bore shall be in accordance with MIL-C-20218, or, the ram shall be provided with a bearing to maintain ram-cylinder separation.

TABLE II. Piles.

Type of piling	Size of piling	Size of hammer					
		A	B	C	D	E	F
Timber	7-in. tip, 12 - to 15-in. butt, 40 ft. long	X					
	Up to 15-in. butt, 9-in. tip and 10 - by 10-in. square piling, 40 ft. long				X		

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

TABLE II. Piles. (Continued)

Type of piling	Size of piling	Size of hammer					
		A	B	C	D	E	F
	7-in. tip, 13 - to 17-in. butt, 60 ft. long		X	X		X	
	Up to 18 in., 14-in. tip and square piling up to 14 - by 14-in. size, 60 ft. long			X		X	
	Up to 24-in., 18-in. tip and square piling up to 24 - x 24- in. size, 60 ft. long						X
Seamless steel pipe	10 in. od, No. 7 gauge, 25 ft. long	X					
	Up to 10 in. od, No. 7 gauge, 25 ft. long				X		
	12 in. od, No. 7 gauge, 60 ft. long		X	X		X	
Concrete and steel-wide flange beam	8 in. and 10 in., 40 ft. long	X					
	10 in. and 12 in., 60 ft. long		X	X		X	
	12 in. and 14 in., 60 ft. long						X
Wide flange beam	8-in., 10-in., and 12-in. sizes, 40 ft. long			X		X	
	10-in., 12-in. sizes, 60 ft. long			X		X	
	12-in., 14-in. sizes, 60 ft. long						X
* Sheet steel piling	3/8-in. web thickness, 40.5 lb. lineal ft., 20 ft. long						
	3/8-in. web thickness, 40.5 lb. lineal ft., 60 ft. long						
Steel pipe and round Concrete pile	Up to 14-in. dia., 60 ft. long			X		X	
	Up to 20-in. dia., 60 ft. long						X

* Sheet steel piling conforming to ASTM A 328.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

3.14 Fuel system. The fuel system shall be complete with all operating accessories, including a pump, tank, valve, filter, hoses, piping, nozzle, and fittings. The tank shall have sufficient capacity for not less than 8 hours of normal operation. The hammer shall operate on type and grade fuel as specified (see 6.2).

3.15 Lubricating system. The lubricating system shall be complete with an oil tank, filter, valves, pumps, hoses, and fittings. The oil tank shall have capacity to lubricate the hammer for not less than 8 hours of normal operation.

3.16 Lifting device, tripping pendent line, and starting device. The hammer shall be equipped with tripping device. If the manufacturer's standard arrangement is designed so that the hammer can be lifted independent of the tripping pendent line when the tripping pendent line is connected to the starting device, the lifting lines or lifting chains conforming to OSHA 1910.184 and a tripping pendent line shall be furnished with the equipment. If the manufacturer's standard arrangement is designed so that the tripping device is used for lifting the hammer as well as for tripping the hammer with a single line, the tripping device design shall allow for easy removal and replacement of the crane's loadline. The starting device shall provide a means of lifting the ram to the top of the cylinder and then releasing the ram automatically to begin its initial starting compression stroke. Engagement of the starting device shall be controlled manually. The hammer shall start in any ambient temperature from -25°F through +125°F. The use of priming agents as an extreme cold temperature initial starting aid is permissible. With pile set and ready for driving, the hammer shall start continuous firing, without the aid of ether or other priming methods, at normal ambient temperature above 32°F as follows:

a. In soft soils, not more than 10 drops of the ram shall be required.
b. In medium hard, clay-type soils, not more than five drops of the ram shall be required.
c. In hard packed, adobe-clay type soils, not more than two drops of the ram shall be required.

3.17 Fuel control system. The hammer shall be furnished with a manual throttle control system which permits the energy output to be adjusted while the hammer is running. There shall also be a back-up fuel shut off method which is independent of the manual throttle. When specified (see 6.2), a remote fuel control system shall be furnished and powered hydraulically, pneumatically, or electrically. The remote control system shall permit the operator, when seated in the crane cab or standing on the ground, to stop the hammer or vary its performance from idle to full power. The remote system shall include 100 feet of control line in increments of 25 feet. When the remote control system is hydraulic- or pneumatic-powered, the control line shall be provided with self-sealing, quick-disconnect couplings. When electrically powered, the remote control system shall be completely waterproof. The control system shall allow for continuous operation of the hammer at any desired throttle setting. The hammer shall shut off automatically in case of failure of either control system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

3.18 Energy output gauge. The hammer shall be equipped with either a mechanical energy output gauge with visible indication of the energy output and energy output chart, or a mechanical stroke measurement gauge with visible indication of the stroke and a chart which correlates stroke to energy output. When specified (see 6.2), closed-end hammers shall be provided with a remote energy output gauge. The gauge shall provide a fast reading of the output energy during driving of the measured air pressure above the ram and in the compression chamber. A 50-foot hose shall be provided for connection of the gauge to the compression chamber with means of shutting off the hose at the gauge. A reference energy chart showing the equivalent weight-height (WH), stored, and potential energy in ft-lb, as obtained from the gauge dial reading, shall be furnished. The gauge and chart shall be enclosed in a metal box with overlapping cover not less than 16 gauge (0.0598 revised United States Standard gauge).

3.19 Leads. Unless otherwise specified (see 6.2), one set of extended commercial standard box-type piledriver leads shall be furnished. The length of the leads shall require a 5-feet headroom clearance, plus normal operating length at maximum energy for the hammer and drive cap, plus the longest length of pile for the size hammer specified in table II. The leads shall be made in sections for bolting or pinning together. All lead sections shall be precision-manufactured in production jigs and top to bottom on each section shall be connected to provide complete field interchangeability between sections. The leads shall be equipped with head blocks and movable boom point connectors with swivel and necessary sheaves and other accessories normally supplied to make a complete workable lead. The boom point connector shall be movable from not greater than 1/3 of the total lead length or to a maximum of 25 feet. The boom point connector for type I hammer shall have a nominal 2.5-inch shaft and for type II hammer a nominal 3-inch shaft. The lead sections shall be furnished with a ladder.

3.19.1 Bottom brace. A bottom brace shall be furnished with accessories for attachment to the crane and lead. The bottom brace shall be adjustable from a minimum length of not less than 15 feet to a maximum length of not less than 25 fee with a minimum right and left side swing of $11^{o} \pm 1^{o}$ including angle from the centerline of the brace at minimum length. The bottom brace adjustment of length and swing shall be changed by mechanical means or hydraulic-driven means as specified (see 6.2). When the bottom brace is hydraulic-driven, means shall be provided to power the hydraulic system.

3.19.2 Leads for tests. Unless otherwise specified (see 6.2), the commercial box leads procured with the hammer shall be used for the performance of the tests in section 4.

3.20 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color and pattern of the finish coat shall be as specified (see 6.2), conforming to FED-STD-595. Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. When specified (see 6.2), the unit shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.21 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be properly lubricated in accordance with manufacturer's standard practice. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.22 Instruction plates. The hammer shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. The marking shall be applied in accordance with MIL-P-514. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.23 Identification marking. Identification shall be permanently and legibly marked directly on the hammer or on a corrosion-resisting metal plate securely attached to the hammer at the source of manufacture. Plates shall be in accordance with MIL-P-514. Identification shall include the manufacturer's model and serial number, name, energy output, ram weight, single or double acting, and trademark to be readily identifiable to the manufacturer. The identification plate shall be securely affixed to the hammer.

3.24 Workmanship. 3.24.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.24.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.24.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.24.4 Welding. Welding procedures shall be in accordance with a AWS D1.1 structural welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.24.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.25 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete hammer when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4.1. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall be conducted on each hammer, leads, and accessories. The inspection shall include the examinations of 4.3, the tests of 4.4.2, and the packaging inspection of 4.5.

4.3 Examination. Each hammer shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard products to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article hammer shall be tested as specified in 4.4.1 and each production hammer shall be subjected to the operational test specified in 4.4.2 to verify satisfactory performance and compliance with the requirements of this specification. Failure to pass any test shall constitute cause for rejection.

4.4.1 First article tests. The ram of the first article hammer shall be weighed. The weight shall be recorded for use in calculating the striking energy. Hammer shall be completely assembled and serviced, then weighed to ascertain that the operating weight does not exceed the maximum specified in 3.9. During the assembling and servicing phase, the hammer shall be observed to verify conformance to the accessibility requirements for repairing, maintaining, and lubricating as specified in 3.7 and 3.21. The first article hammer shall be subjected to the tests specified in 4.4.1.1 through 4.4.1.4, and shall be run on the type and grade fuel as specified (see 6.2). Failure to pass one or more tests shall be cause for rejection.

4.4.1.1 Piledriving. Leads as specified in 3.19.2 shall be used during this test. Test shall be performed at normal ambient temperature within the limits as specified in 3.10. The contractor shall furnish one of each type of piles conforming to table II that is applicable to the size of hammer being tested. When specified (see 6.2), piles shall be driven their full length or to refusal into each type of soil as described in 3.16. Throughout the test, hammer shall be observed to determine conformance to the starting requirements of 3.16 and the fuel, lubricating, and control systems of 3.14, 3.15, and 3.17 respectively.

4.4.1.2 Refusal. Hammer shall be mounted on a solid block, a pile that has been driven to refusal, or a test stand that simulates driving at refusal. Hammer shall be operated for not less than 50 hours. Only normal maintenance and scheduled servicing shall be performed on the hammer during this test. Throughout the test, hammer shall be observed to determine compliance with 3.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

4.4.1.3 Energy. The hammer shall be mounted on the block, pile, or test stands used in 4.4.1.2. Hammer shall be started and adjusted until operating at a constant cyclic rate of not less than 36 strokes per minute. A stroke indicator shall be used to measure the length of stroke. The length of stroke shall be measured by averaging any 10 consecutive strokes at the constant cyclic rate specified herein. To verify compliance with the requirements of table I, the striking energy shall be calculated using the formula specified in 3.9.

4.4.1.4 Horsepower. When the first article hammer is size D or E, the hp shall be calculated using the formula specified in 3.9 to prove conformance to the requirements of table I.

4.4.2 Operational test. Each hammer shall be completely assembled, properly adjusted, serviced for operation, and operated for not less than 45 minutes. At least 5 minutes of this operation shall be at maximum striking energy of the hammer at refusal. Hammer shall either drive piles or operate on a block, a pile driven to refusal, or a test stand. All controls shall be operated not less than 12 times or as many times as necessary to determine ease of operation, effectiveness, responsiveness, and that mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction. Hammer shall be actually examined for excessive heating and observed for abnormal operation. Inability to operate for 45 minutes, evidence of any malfunction, or failure to pass any phase of this test shall be cause for rejection.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss and to accomplish reduction in cube. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed and mating parts on the equipment shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as in 5.1.1.6 and 5.1.1.7.

5.1.1.6 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.7 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2 or MIL-B-22191, type II. The material shall be secured in place with tape conforming to PPP-T-76.

5.1.1.8 Cylinder and ram. All interior surfaces of the cylinder and ram shall be coated with type P-10 preservative. The preservative shall be applied through the lubricator by actuating the ram with the fuel turned off. In addition, the combustion area shall be thoroughly coated with the preservative by fogging through the air intake port. Openings to cylinder shall be sealed with tape conforming to PPP-T-60, type IV or type III, class I.

5.1.1.9 Fuel and lubrication oil tanks. The tanks shall be drained and the interior surfaces coated with type P-10 preservative.

5.1.1.10 Maintenance tools. Maintenance tools shall be preserved in accordance with the level a preservation and packaging requirements of PPP-P-40.

5.1.1.11 Technical publications. Technical publications for each hammer shall be preserved, method IC-1 or IC-3.

5.1.1.12 Consolidated packaging. Components and parts removed from each hammer, accessories, maintenance tools, and technical publications be packaged in close-fitting boxes conforming to PPP-B-621, class 1; or shall PPP-B-601, domestic type. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes.

5.1.2 Commercial. The equipment shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Each complete hammer together with packaged component parts, tools, and publications shall be packed in a crates conforming to MIL-C-104, type II. When specified (see 6.2) crates shall be type I. The contents shall be anchored, blocked, and braced in accordance with the crate specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Each complete hammer together with packaged component parts, tools, and publications shall be packed in an open type crate conforming to MIL-C-3774, type II. When specified (see 6.2), crates conforming to MIL-C-3774, type I shall be used. The contents shall be anchored, blocked, and braced in accordance with the crate specification.

5.2.3 Commercial. The equipment shall be prepared for shipment in accordance with ASTM D 3951. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The piledriver hammer with leads and bottom brace is intended to be used with a crane having a lattice-type boom for driving piles of the maximum length specified in table II for the size of hammer listed. Longer length piles may be driven by splicing a new section of pile onto the driven pile. However, the maximum driven pile length should not exceed a ram weight to pile weight ratio of 1 to 4. Both side by side and fore and aft batter piles may be driven with a ratio of the vertical rise per unit of horizontal affect of 3 to 1 maximum.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specifications.
b. Type, size of hammer required (see 1.2).
c. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. When lead guides are not required (see 3.11).
e. Type of drive caps required (see 3.12).
f. Type and quantity of cushion block required (see 3.12).
g. When a remote fuel control system is required (see 3.17).
h. When remote energy output gauge is required on hammer (see 3.18).
i. When leads are to be different (see 3.19).
J. When bottom brace is to be adjustable by mechanical means or hydraulic driven (see 3.19.1).
k. When leads used for tests shall be different (see 3.19.2).
l. When testing in different soils is required (see 4.4.1.1).
m. Type of fuel required for operation, tests, and inspections (see 3.14, 4.21, and 4.22).
n. Color and pattern of finish coat required (see 3.20).
o. When hammer must be cleaned, treated, and painted in accordance with MIL-T-704 (see 3.20).
p. Level of preservation and packing required (see 5.1 and 5.2).
q. When type I crates are required (see 5.2.1 and 5.2.2).

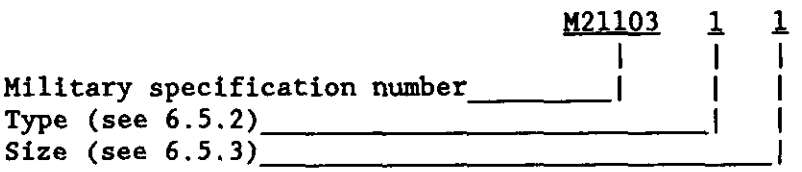
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.3), and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for sample for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Part or identifying number (PIN). The PIN corresponds to the type and size data applicable to units covered by this specification and defines the requirements of the options presented under this specification. The military specification number and the type and size code numbers are combined to form the PIN.

6.5.1 Cataloging data. For cataloging purposes, PIN for the units are assigned as follows:



6.5.2 Type. The type of the hammer (see 1.2) is identified by a single numerical character (see table III).

TABLE III. Code number to type.

Type	Code
I	1
II	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

6.5.3 Size. The size of the hammer (see 1.2) is identified by single alphabetic character (see table IV).

TABLE IV. Code letter to size.

Size	Code
A	1
B	2
C	3
D	4
E	5
F	6

6.6 Subject term (key word) listing:

Construction equipment
Impactors

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD

Preparing activity:
Navy - YD

Review activity:
Navy - MC

Project No. 3895-0294

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-21103E	MILITARY SPECIFICATION HAMMER, PILEDRIVER, SELF-POWERED, DIESEL

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:54Z
Check the source to verify that this is the current version before use.